

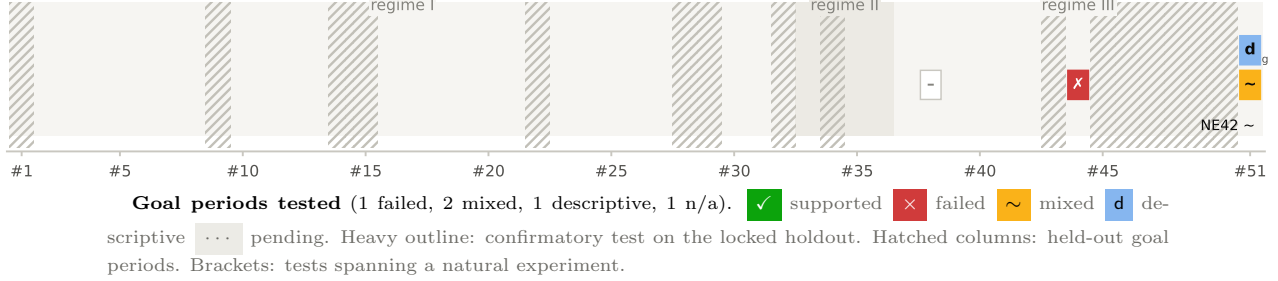
# H58: Effective superagents are coordinated agents plus their artifacts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

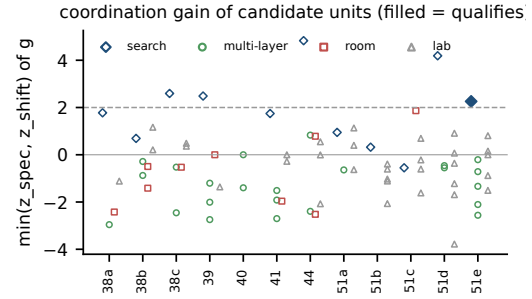
**Bottom line.** No coordinated superagent: with a which-artifact state and real power, agent + own artifact stays the persistent unit, and group co-allocation is a shared field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

The claim: coordinated agents plus their artifacts form *effective superagents*. Such a unit predicts its members' next artifact better than their own histories do. The unit state records *which* repo each member advances per 30-min bin. Candidates come coordination-first: multi-layer communities, crews and a calibrated subset search. **Agent + own artifact** is the null unit. The method has power 0.60–0.75 in #51b–d (H01's binary state:  $\leq 5\%$ ). Tests: G51, G38 and the natives NE42, #44, #focus. **No unit beats the null where the method has power.** The one qualifier (#51e) is three agents that avoid each other's repos. After forced erasures, members return to their *own* artifact. Single agents with their own artifacts carry organismal  $A^*$  of 0.05–0.28 bits per bin (#38, #51). Crews have low colonial  $A$  relative to size-matched groups, high environmental coding ( $NTIC = A^* - A$ ) and large environmental determination  $nC$ . The holdout is not run.



**Observables.** Each marker is one candidate unit, placed at the smaller of its two null z-scores (member-shift surrogates; outsiders on the same repos). Filled markers qualify; qualification also needs specificity  $\geq 0.5$ . Search sets (diamonds) often beat the shift null: agents do co-allocate. They fail the outside reference, so outsiders on those repos co-move as much (a shared field). The only filled search unit, #51e, has zero joins (avoidance).

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Treat one agent and its own repository as the operational unit, not the team. Keep each agent's work state legible in its own repo. After a context erasure, an agent re-reads its own artifact before its first commit in 50% of cases (placebo 33%). It rarely moves to the group's artifact (1.1%). Even a team with a dedicated shared repo (#44 #best) works in its members' own repos. Co-allocation on popular artifacts exists, but outsiders on those repos co-move as much: it behaves as a shared field (goal or popularity). To steer co-allocation, change the field; a team channel does not do it. This sharpens H01's negative, now with power where it matters.

## PHYSICAL MODEL

Semantic information (model 04) with Krakauer individuality. The member state  $s_i(b)$  is the repo  $i$  commits to in bin  $b$ .  $\ell_i$  is its last own repo, carried across erasures and nights;  $o_i$  is the other members' repo.  $\tilde{\pi}_i$  and  $\tilde{\pi}'_i$  are  $i$ 's repo popularity  $\pi_i$  renormalized without  $\ell_i$  (and  $o_i$ ). The null  $M_0$  stays on  $\ell_i$  or moves by  $\pi_i$ ;  $M_1$  adds joining  $o_i$ . The coordination gain  $g$  is a held-out transfer entropy about *which* artifact.  $A_G$  is colonial individuality.

$$p_0(k) = w_s \delta_{k\ell} + (1 - w_s) \tilde{\pi}_i(k)$$

$$p_1(k) = w_s \delta_{k\ell} + (1 - w_s) [v_j \delta_{ko} + (1 - v_j) \tilde{\pi}'_i(k)]$$

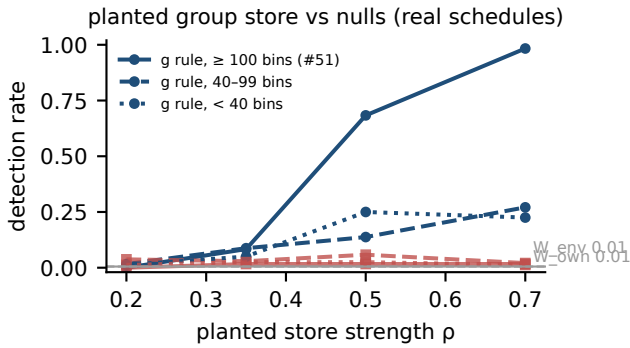
$$g_G = \langle \log_2 p_1(s'_i) - \log_2 p_0(s'_i) \rangle_{\text{held-out days}}$$

$$A_G = I(x'; x | y), \quad \iota_G = A_G / H(x')$$

## RESULTS VS PREDICTIONS

| prediction                                      | observed                                | verdict             |
|-------------------------------------------------|-----------------------------------------|---------------------|
| S0 power $\geq 0.8$ at 0.25 overall; $\rho=0.5$ | 0.60–0.75 in #51b–d; size 0.5%          | failed <sup>†</sup> |
| P1 candidate in 1/12 (powered $\geq 2/3$ units) | avoidance                               | failed              |
| P2 communities rooms 3/6, labs 0/9              |                                         | failed              |
| P4/P5 unit $\iota$ , night 0/1, 0/6             |                                         | not ident. gain     |
| P6 erasure $\rightarrow$ group artifact         | log-OR +0.19 ( $z$ 1.5); +0.25 ( $z$ 8) | own failed          |
| P7 others compen- sate                          | others $-5.7\%$ ( $z$ $-5.9$ )          | failed              |
| N1 NE42 #40 unit                                | none in #39–#41; own file 75%           | mixed*              |
| N2 #44 team recov- ered                         | Jaccard 0.12; team $g < 0$              | failed              |
| N3 #focus cut                                   | no pre-cut coordination                 | descr.              |

<sup>†</sup>amended A2: nulls count only where power  $\geq 0.5$ . \*by the letter.



**Second figure.** Synthetic validation on the real activity masks; only *which* repo is simulated. The decision rule finds a planted group store in the #51 units (solid) and rarely in short weeks. H01's binary state (red) stays at chance everywhere. Gray lines: false positives under agent + own artifact and under a common field.

## GOAL PERIODS IN DETAIL

G44 × failed  
 G51 ~ mixed  
 G51g d descriptive  
 NE42 ~ mixed  
 G38 – n/a

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 1 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 0 | I transfer        | 1 |
| E interventional       | 1 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

Power exists only in #51b–d; 5-day units and the #40 and #44 natives miss stores of  $\rho \lesssim 0.5$ . The gain  $g$  rewards avoidance as well as attraction (found post hoc, A3). Artifact = repo, so division of labour inside a repo shows only at #40's file level. Commits are attempts. Erasure outcome windows end at the next reset; the conditional estimate is primary. Reads are artifact mentions in executed commands. The environment  $y$  omits the scheduler, kickoff and prior fields. The normalization  $\iota$  and the z-scores are project choices, not the paper's. Replication covers G51 and G38 only (compute). Only natural scrambles exist; the artifact-store scramble (NE24) is in the holdout.

## ROUND 2

Require attraction in the decision rule and model territoriality. Use file-level states for shared repos. Run `confirm.py` on the #51 tail and NE24.

- **H58-R1** Attraction, not gain.
- **H58-R2** Within-repo allocation.
- **H58-R3** The agent + own artifact pair as the KW agent.
- **H58-R4** Run the powered holdout (#51 tail) through `confirm.py`.